

TREMORTRACE · DESIGN PRECIS

TremorTrace design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/tremortrace		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

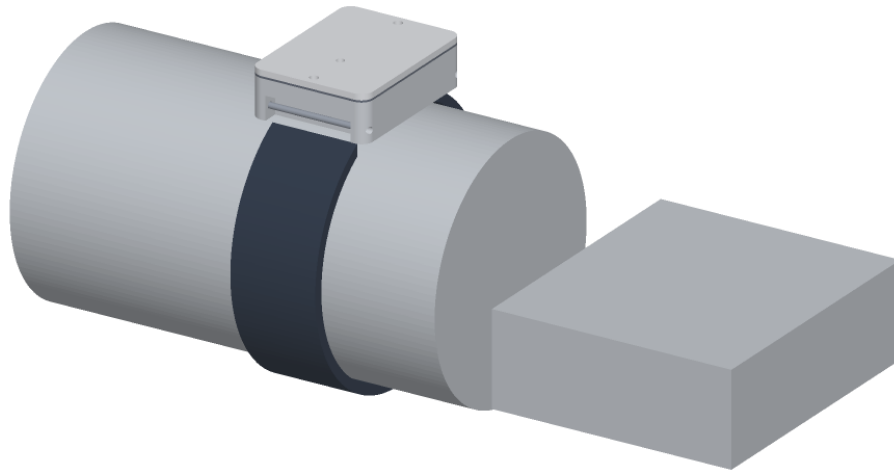
Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	TRL 3: decisions from TRT-DDR-001 recorded; numbers from TRT-CAL-001 (storage ratio 390, not 1,000; firmware in internal flash; mass 29.4 g); parametric model and TRT-DWG-002
0.3	2026-09-24	Amish Chadha	Review fixes: one-module and local-first choices marked proposed; massing pod closed at 12 mm; concept sheet renumbered TRT-DWG-001
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

TremorTrace design precis

TremorTrace is a small wrist pod on a watch strap that records wrist motion all day, turns it into 10-second tremor summaries on the band, and hands them to an open notebook that draws a daily tremor profile. The TRL 3 calculations (TRT-CAL-001) show that one off-the-shelf module and a 150 mAh cell meet nine of eleven requirements for \$41.19 in parts; mass is at risk at 29.4 g against a 30 g limit, and splash resistance and the notebook can only be verified with hardware and software work.

TremorTrace

CONCEPT, NOT FOR FABRICATION



Grey: Forearm and hand for scale

■ How it works

1. **Sense.** A 6-axis IMU (accelerometer and gyroscope) samples wrist motion at 104 Hz, with ranges of ± 8 g and ± 2000 dps.
2. **Summarize on the band.** Every 10 seconds the firmware band-pass filters the signal to 3 to 15 Hz, runs a spectrum, and stores a 32-byte summary: dominant frequency, band RMS of acceleration and angular velocity, total RMS, a band-to-low-frequency power ratio and an activity flag. Tremor values are kept even when the flag says the wearer was moving, so tremor during activity is not lost.
3. **Store.** Summaries go to the module's 2 MB flash, enough for 9.7 days at 16 h of wear per day without a phone. A 256 kB area holds up to 21 raw 10 s snippets captured on demand for research.

4. **Sync.** When the owner opens the web Bluetooth page or a laptop script, summaries transfer over Bluetooth Low Energy. Nothing goes to the cloud.
5. **Analyze.** An open Python notebook turns summaries into a daily profile: minutes of tremor per hour, dominant frequency and amplitude over the day, and optional medication-time markers entered by the owner.

TremorTrace: data flow (local-first, no cloud)

CONCEPT, NOT FOR FABRICATION



■ Main components

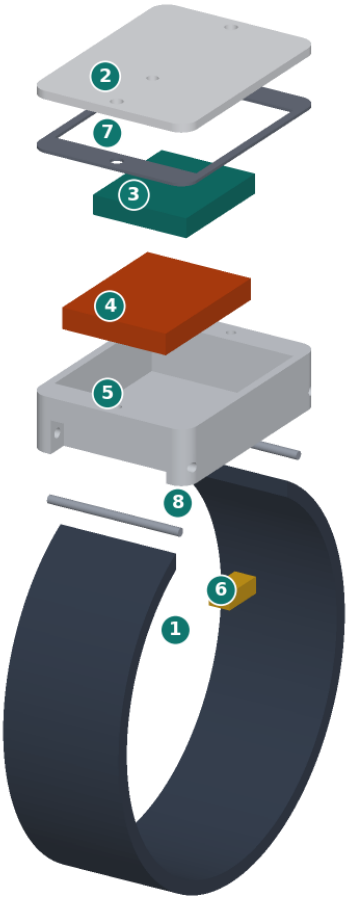
Table 1. Main components; item numbers match `bom/bom.csv` and the exploded view.

#	Component	Choice	Notes
1	Strap	22 mm silicone quick-release watch strap	Skin-safe; the heaviest part (about 12 g, assumed)
2	Enclosure lid	3D-printed PETG, 1.5 mm, two M2 screws	2 mm light pipe over the module's status LED
3	Controller and IMU	Seeed XIAO nRF52840 Sense class: nRF52840, LSM6DS3TR-C IMU, 2 MB flash, BQ25101 charger	One module, no custom PCB. Decided by Amish, 2026-09-25 (TRT-DDR-001)
4	Battery	150 mAh protected LiPo, 19.75 x 26.02 x 3.8 mm (Adafruit 1317 class)	See power budget
5	Enclosure base	3D-printed PETG with integrated 22 mm lugs	Charging connector through the floor
6	Charging contacts	Magnetic 2-pin pogo connector	Keeps the enclosure sealed
7	Gasket	Printed TPU ring, 0.5 mm compressed	Toward the IP54 target (R7)
8	Spring bars	22 mm	Hold the strap in the lugs

TremorTrace: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Silicone strap
- 2 Enclosure lid
- 3 Controller and IMU module
- 4 LiPo cell, 150 mAh
- 5 Enclosure base
- 6 Charging connector
- 7 TPU gasket
- 8 Spring bars



■ **Key numbers**

All values come from TRT-CAL-001 and the parametric model in `cad/src/model.py`.

Table 2. Key numbers at TRL 3.

Quantity	Value	Requirement
Frequency resolution	0.100 Hz bins (10 s at 104 Hz)	R1 (0.2 Hz) met, with sample-rate correction
Nyquist limit	52 Hz	R1, R3 met
In-band noise	0.31 mg and 0.017 dps RMS	R2 met
Recording current	1.12 mA nominal, 1.54 mA conservative	
Battery life	6.3 days nominal, 4.5 days conservative	R4 (3 days) met, if the firmware sleeps
Summary storage	184.3 kB per day (32 bytes per 10 s, 16 h)	
On-band history	9.7 days	R5 (7 days) met
Pod size	40 x 30 x 12 mm	R6 size met
Mass	29.4 g with strap	R6 (30 g) at risk
Parts cost	\$41.19	R10 met

The general arrangement is drawing TRT-DWG-002 ([cad/drawings/TRT-DWG-002.pdf](#)).

■ Key design choices

- **Summaries, with optional raw snippets.** A 32-byte summary is 390 times smaller than the 12,480 bytes of raw data in a 10 s window, and it keeps personal data minimal. Decided by Amish, 2026-09-25 (TRT-DDR-001).
- **One module, no custom PCB.** Keeps the first build within reach of anyone with a soldering iron and a 3D printer. Decided by Amish, 2026-09-25 (TRT-DDR-001).
- **Web Bluetooth page first.** No app store and no account; a native app may follow. Decided by Amish, 2026-09-25 (TRT-DDR-001).
- **Local-first data.** No cloud account is needed to use or analyze the data (R9). Decided by Amish, 2026-09-25 (TRT-DDR-001).
- **Most affected wrist first.** Decided by Amish, 2026-09-25 (TRT-DDR-001).
- **Charge current 50 mA.** Charges in about 3.6 h with about 65 mW of charger heat, against 1.8 h and 130 mW at 100 mA. Proposed, awaiting Amish.

■ Safety

Safety: Research and educational prototype only. It is not a medical device, has not been cleared or approved by any regulator, and must not be used to diagnose, treat or adjust medication.

- The LiPo cell sits against the skin: use a protected cell, never charge while worn, charge at no more than the cell's 150 mA limit, and stop use if the pod becomes warm or swollen.
- Use skin-safe strap materials and check for irritation during long wear.
- Tremor data is health data. Keep it on the owner's devices and share only with consent.

■ Open questions

- The activity flag marks tremor during a reach as activity in synthetic data (TRT-CAL-001, section 4). How well can action tremor be separated from voluntary movement with 10 s summaries?
- Confirm the IMU sample-rate tolerance, the module's run current and the strap mass.
- First clinical or patient partner for co-design and eventual validation. Proposed, awaiting Amish.

Concept media: [blueprint sheet](#), [interactive 3D model](#).